

Machine Learning week – 1 Assignments

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```
import numpy as np
import pandas as pd
df=pd.read_csv('Data.csv')
x=df.iloc[:, :-1].values
y=df.iloc[:, -1].values
```

#task-1

```
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(missing_values=np.nan, strategy='median')
x[:, 1:3]=imputer.fit_transform(x[:, 1:3])
print(x)
```

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
ct=ColumnTransformer(transformers=[('encoder', OneHotEncoder(), [0])], remainder='passthrough')
x=np.array(ct.fit_transform(x))
print(x)
```

```
from sklearn.preprocessing import LabelEncoder
lb=LabelEncoder()
y=lb.fit_transform(y)
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x=sc.fit_transform(x)
np.set_printoptions(suppress=True)
print(x)
```

#task-2

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=1)
print(x_train)
```

R²_SCORE: *(coefficient of determination regression score function)*

$$\text{R-Square} = \frac{\sum(Y_{\text{actual}} - Y_{\text{predicted}})^2}{\sum(Y_{\text{actual}} - Y_{\text{mean}})^2}$$

- *r²_score is imported from sklearn.metrics.*
- *r² is a statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable(s) in a regression model.*
- *r² explains to what extent the variance of one variable explains the variance of second variable.*

Calculating R²:

The actual calculation of R-squared requires several steps.

- *Taking the data points (observations) of dependent and independent variables and finding the line of best fit, often from a regression model.*
- *Calculate predicted values, subtract actual values and square the results. This yields a list of errors squared, which is then summed and equals the unexplained variance.*
- *To calculate the total variance, subtract the average actual value from each of the actual values, square the results and sum them. From there, divide the first sum of errors (explained variance) by the second sum (total variance), subtract the result from one.*

In this way we get the r²value.

$$R^2 = 1 - (\text{Total Variation}) / (\text{Unexplained Variation})$$

- *r² takes the values between 0 and 1.*
- (It can also be negative when the model can be arbitrarily worse)*

Name:

Soumya Moturi

College:

Gudlavalleru Engineering College